Solving for the Minimum Mean-Squared Error Weights

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Key Concepts

1) The generic minimum mean-squared error (MMSE) filtering problem defines an error signal

$$e[n] = d[n] - \sum_{k=0}^{N-1} w_k x[n-k]$$

where w_k are the coefficients of an FIR filter, $d[n]$ and $x[n]$ are input signals.

2) The MMSE filter design problem chooses w_k to minimize a function of the squared error for a given set of signals $d[n]$ and $x[n]$.

Let $\mathbf{w} = \begin{bmatrix} w_0 & w_1 & \dots & w_{N-1} \end{bmatrix}^T$ be an N-by-1 vector of FIR filter weights.

a) Statistical MMSE

$$\min_{\mathbf{w}} \mathsf{E}\left\{e^2[n]\right\}$$

b) Practical MMSE

$$\min_{\mathbf{w}} \frac{1}{L} \sum_{n=l_1}^{l_1+L-1} e^2[n]$$

c) We will derive the solution for the practical MMSE cost function and show that the solution for the statistical MMSE problem takes similar form.

3) Use a vector-matrix formulation of the problem.

   a) Boldface (underlined in video) symbols denote vectors and matrices.

   b) Superscripts T and -1 denote vector/matrix transpose and matrix inverse, respectively.

   c) If $\mathbf{a}$ and $\mathbf{b}$ are column vectors, then $\mathbf{a}^T\mathbf{b} = \mathbf{b}^T\mathbf{a}$.

   d) If $\mathbf{A}$ is a matrix and $\mathbf{b}$ a vector, then $(\mathbf{A}\mathbf{b})^T = \mathbf{b}^T\mathbf{A}^T$.

4) Write error in terms of the inner product of N-dimensional vectors

$\mathbf{w}$ and $\mathbf{x}^T[n] = \begin{bmatrix} x[n] & x[n-1] & \dots & x[n-N+1] \end{bmatrix}$

$$e[n] = d[n] - \mathbf{x}^T[n]\mathbf{w}$$

5) Now collect errors into a L-by-1 vector $\mathbf{e}$ so the MMSE filtering problem is

$$\min_{\mathbf{w}} \frac{1}{L}\mathbf{e}^T\mathbf{e}$$

where

$$\mathbf{e} = \begin{bmatrix} e[l_1] \\ e[l_1+1] \\ \vdots \\ e[l_1+L-1] \end{bmatrix} = \begin{bmatrix} d[l_1] \\ d[l_1+1] \\ \vdots \\ d[l_1+L-1] \end{bmatrix} - \begin{bmatrix} \mathbf{x}^T[l_1] \\ \mathbf{x}^T[l_1+1] \\ \vdots \\ \mathbf{x}^T[l_1+L-1] \end{bmatrix} \mathbf{w}$$

To simplify notation we write this as

$$\mathbf{e} = \mathbf{d} - \mathbf{X}^T\mathbf{w}$$

where the L-by-1 vector $\mathbf{d} = \begin{bmatrix} d[l_1] & d[l_1+1] & \cdots & d[l_1+L-1] \end{bmatrix}^T$ and the L-by-N matrix

$$\mathbf{X}^T = \begin{bmatrix} \mathbf{x}^T[l_1] \\ \mathbf{x}^T[l_1+1] \\ \vdots \\ \mathbf{x}^T[l_1+L-1] \end{bmatrix}$$

6) Expand the MMSE to write it as a function of $\mathbf{w}$

$$\begin{aligned}\frac{1}{L}\mathbf{e}^T\mathbf{e} &= \frac{1}{L}\left(\mathbf{d}-\mathbf{X}^T\mathbf{w}\right)^T\left(\mathbf{d}-\mathbf{X}^T\mathbf{w}\right)\\ &= \frac{1}{L}\left(\mathbf{d}^T-\mathbf{w}^T\mathbf{X}\right)\left(\mathbf{d}-\mathbf{X}^T\mathbf{w}\right)\\ &= \frac{1}{L}\mathbf{d}^T\mathbf{d}-2\frac{1}{L}\mathbf{d}^T\mathbf{X}^T\mathbf{w}+\mathbf{w}^T\left(\frac{1}{L}\mathbf{X}\mathbf{X}^T\right)\mathbf{w}\\ &= \sigma_d^2-2\mathbf{p}^T\mathbf{w}+\mathbf{w}^T\mathbf{R}\mathbf{w}\end{aligned}$$

where $\sigma_d^2=\frac{1}{L}\mathbf{d}^T\mathbf{d}$, $\mathbf{p}^T=\frac{1}{L}\mathbf{d}^T\mathbf{X}^T$ and $\mathbf{R}=\left(\frac{1}{L}\mathbf{X}\mathbf{X}^T\right)$.

7) Side comments:

a) Note that

$$\sigma_d^2 = \frac{1}{L} \sum_{n=l_1}^{l_1+L-1} d^2[n]$$

$$\mathbf{p} = \frac{1}{L} \sum_{n=l_1}^{l_1+L-1} d[n]\mathbf{x}[n]$$

$$\mathbf{R} = \frac{1}{L} \sum_{n=l_1}^{l_1+L-1} \mathbf{x}[n]\mathbf{x}^T[n]$$

are L-sample estimates of the variance E$\{d^2[n]\}$, N-by-1 cross correlation vector E$\{d[n]\mathbf{x}[n]\}$, and N-by$N$ correlation matrix E$\{\mathbf{x}[n]\mathbf{x}^T[n]\}$.

b) Since here we are interested in $L > N$ and in signal pro-

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cessing noise is always present, the matrix $\mathbf{R}$ is invertible. We will not prove this.

8) Use the method of completing the square to solve

$$\min_{\mathbf{w}} \sigma_d^2 - 2\mathbf{p}^T\mathbf{w} + \mathbf{w}^T\mathbf{R}\mathbf{w}$$

This is accomplished by writing

$$\sigma_d^2 - 2\mathbf{p}^T\mathbf{w} + \mathbf{w}^T\mathbf{R}\mathbf{w} = (\mathbf{w} - \mathbf{w}_o)^T\,\mathbf{R}\,(\mathbf{w} - \mathbf{w}_o) + \sigma_d^2 - \mathbf{p}^T\mathbf{R}^{-1}\mathbf{p}$$

where $\mathbf{w}_o = \mathbf{R}^{-1}\mathbf{p}$. Substitution for $\mathbf{w}_o$ and multiplication of terms verifies the right and left hand sides of this equation are equal.

9) Note that

$$\begin{aligned}\mathbf{a}^T\mathbf{R}\mathbf{a} &= \frac{1}{L}\sum_{n=l_1}^{l_1+L-1}\mathbf{a}^T\mathbf{x}[n]\mathbf{x}^T[n]\mathbf{a} \\ &= \frac{1}{L}\sum_{n=l_1}^{l_1+L-1}\left(\mathbf{a}^T\mathbf{x}[n]\right)^2 \\ &\geq 0\end{aligned}$$

for any $\mathbf{a}$.

10) Hence, the minimum value of the mean-squared error is

$$\min_{\mathbf{w}}\left(\mathbf{w}-\mathbf{w}_o\right)^T\mathbf{R}\left(\mathbf{w}-\mathbf{w}_o\right)+\sigma_d^2-\mathbf{p}^T\mathbf{R}^{-1}\mathbf{p}=\sigma_d^2-\mathbf{p}^T\mathbf{R}^{-1}\mathbf{p}$$

and is obtained by choosing the optimum filter weights as

$$\mathbf{w} = \mathbf{w}_o = \mathbf{R}^{-1}\mathbf{p}$$

11) *Connection to Least Squares.* Minimizing $\mathbf{e}^T\mathbf{e}$ implies we seek $\mathbf{w}$ that satisfies

$$\mathbf{X}^T\mathbf{w} \approx \mathbf{d}$$

This represents a system of L linear equations with N unknowns. Here we are interested in the case where $L > N$.

a) Multiply both sides of the equation by $\mathbf{X}$ to form the normal equations

$$\mathbf{X}\mathbf{X}^T\mathbf{w} = \mathbf{X}\mathbf{d}$$

b) Now invert $\mathbf{X}\mathbf{X}^T$ to obtain

$$\begin{aligned}\mathbf{w} &= \left(\mathbf{X}\mathbf{X}^T\right)^{-1}\mathbf{X}\mathbf{d} \\ &= \mathbf{R}^{-1}\mathbf{p}\end{aligned}$$

The MMSE filtering problem is equivalent to the least squares problem involving $L > N$ equations in N unknowns.

12) *Connection to Statistical MMSE Filtering.* The statistical MMSE problem is of the form

$$\min_{\mathbf{w}} \mathsf{E}\{e^2[n]\} = \min_{\mathbf{w}} \mathsf{E}\left\{\left(d[n] - \mathbf{w}^T\mathbf{x}[n]\right)\left(d[n] - \mathbf{x}^T[n]\mathbf{w}\right)\right\}$$

Multiply terms and take the expectation to obtain

$$\min_{\mathbf{w}} \sigma_d^2 - 2\mathbf{p}^T\mathbf{w} + \mathbf{w}^T\mathbf{R}\mathbf{w}$$

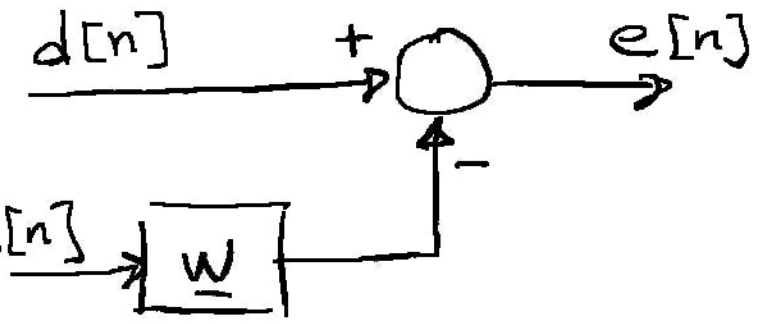
where $\sigma_d^2 = \mathrm{E}\{d^2[n]\}$, $\mathbf{p} = \mathrm{E}\{d[n]\mathbf{x}[n]\}$, and $\mathbf{R} = \mathrm{E}\{\mathbf{x}[n]\mathbf{x}^T[n]\}$. This has the same form as the practical MMSE problem solved above, so we know that the solution is $\mathbf{w} = \mathbf{R}^{-1}\mathbf{p}$.

13) *Structure of Data Matrix* $\mathbf{X}$. Substituting the definition of $\mathbf{x}[n]$ into the definition of $\mathbf{X}$ gives

$$\mathbf{X} = \begin{bmatrix} x[l_1] & x[l_1+1] & \cdots & x[l_1+L-1] \\ x[l_1-1] & x[l_1] & \cdots & x[l_1-L-2] \\ x[l_1-2] & x[l_1-1] & \cdots & x[l_1+L-3] \\ \vdots & \vdots & & \vdots \\ x[l_1-N+1] & x[l_1-N+2] & \cdots & x[l_1+L-N] \end{bmatrix}$$

This matrix has the same value along every diagonal and thus is said to be *Toeplitz*.

MMSE Filtering



$$e[n] = d[n] - \sum_{k=0}^{N-1} w_k x[n-k]$$

- Objective: minimize mean-squared error

$$\min_{\underline{w}} \frac{1}{L} \sum_{n=\ell_1}^{\ell_1+L-1} e^2[n]$$

- Use matrix-vector formulation

$$\underline{w} = [w_0 \; w_1 \cdots w_{N-1}]^T \qquad \underline{x}^T[n] = [x[n] \; x[n-1] \cdots x[n-N+1]]$$

$$e[n] = d[n] - \underbrace{[x[n] \; x[n-1] \cdots x[n-N+1]]}_{1 \times N} \underbrace{\begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_{N-1} \end{bmatrix}}_{N \times 1} = d[n] - \underline{x}^T[n]\,\underline{w}$$

Put errors into a vector $\underline{e}$

$$\underset{L\times 1}{\underline{e}} = \begin{bmatrix} e[\ell_1] \\ e[\ell_1+1] \\ \vdots \\ e[\ell_1+L-1] \end{bmatrix} = \underbrace{\begin{bmatrix} d[\ell_1] \\ d[\ell_1+1] \\ \vdots \\ d[\ell_1+L-1] \end{bmatrix}}_{\underline{d}\,:\,L\times 1} - \underbrace{\begin{bmatrix} \underline{x}^T[\ell_1] \\ \underline{x}^T[\ell_1+1] \\ \vdots \\ \underline{x}^T[\ell_1+L-1] \end{bmatrix}}_{\underline{X}^T\,:\,L\times N} \underline{w}$$

MMSE

$$\min_{\underline{w}} \frac{1}{L}\sum_{n=\ell_1}^{\ell_1+L-1} e^2[n] \Rightarrow \min_{\underline{w}} \frac{1}{L}\underline{e}^T\underline{e}$$

$$\underline{e}^T\underline{e} = (\underline{d}-\underline{X}^T\underline{w})^T(\underline{d}-\underline{X}^T\underline{w}) = (\underline{d}^T-\underline{w}^T\underline{X})(\underline{d}-\underline{X}^T\underline{w})$$

$$\text{So } \frac{1}{L}\underline{e}^T\underline{e} = \underbrace{\frac{1}{L}\underline{d}^T\underline{d}}_{\sigma_d^2} - 2\underbrace{\left(\frac{1}{L}\underline{d}^T\underline{X}^T\right)}_{\underline{P}^T}\underline{w} + \underline{w}^T\underbrace{\left(\frac{1}{L}\underline{X}\underline{X}^T\right)}_{\underline{R}}\underline{w}$$

Rewriting $\min_{\underline{w}} \left(\sigma_d^2 - 2\underline{p}^T\underline{w} + \underline{w}^T\underline{R}\underline{w} \right)$

where $\sigma_d^2 = \frac{1}{L}\sum_{n=l_1}^{l_1+L-1} d^2[n]$; $\underline{p} = \frac{1}{L}\sum_{n=l_1}^{l_1+L-1} \underline{x}[n]d[n]$; $\underline{R} = \frac{1}{L}\sum_{n=l_1}^{l_1+L-1} \underline{x}[n]\underline{x}^T[n]$

Another view $\min_{\underline{w}} \frac{1}{L}\underline{e}^T\underline{e}$; $\underline{e} = \underline{d} - \underline{X}^T\underline{w} \Rightarrow \underline{d} \approx \underline{X}^T\underline{w}$

L equations, N unknowns

Three possibilities

Assuming noise is present

1) $L < N$ nonunique solution for $\underline{e} = \underline{0}$
2) $L = N$ unique solution possible for $\underline{e} = \underline{0}$
3) $L > N$ no solution

We are generally interested here in $L > N$

Solving for $\underline{w}$: use "completing the square"

- Scalar case $\min_w \sigma_d^2 - 2pw + rw^2 \Rightarrow r(w - r^{-1}p)^2 + \sigma_d^2 - p^2 r^{-1}$

  if $r > 0$, then $w = r^{-1}p$ is the solution

- Vector case

  rewrite $\sigma_d^2 - 2\underline{p}^T\underline{w} + \underline{w}^T\underline{R}\underline{w} = (\underline{w} - \underline{w}_o)^T\underline{R}(\underline{w} - \underline{w}_o) + \sigma_o^2$

  1) We need $\underline{p}^T\underline{w} = \underline{w}_o^T\underline{R}\underline{w}$ or $\underline{p}^T = \underline{w}_o^T\underline{R} \Rightarrow \underline{w}_o = \underline{R}^{-1}\underline{p}$

  2) Now $(\underline{w} - \underline{R}^{-1}\underline{p})^T\underline{R}(\underline{w} - \underline{R}^{-1}\underline{p}) = \underline{w}^T\underline{R}\underline{w} - 2\underline{p}^T\underline{w} + \underline{p}^T\underline{R}^{-1}\underline{p}$

  $\Rightarrow \sigma_o^2 = \sigma_d^2 - \underline{p}^T\underline{R}^{-1}\underline{p}$ independent of $\underline{w}$

  3) Note $\underline{a}^T\underline{R}\underline{a} = \frac{1}{L}\sum_{n=l_1}^{l_1+L-1}(\underline{a}^T\underline{x}[n])(\underline{x}^T[n]\underline{a}) = \frac{1}{L}\sum_{n=l_1}^{l_1+L-1}(\underline{x}^T[n]\underline{a})^2 \geq 0$ for all $\underline{a}$

MMSE: $\underline{w} = \underline{R}^{-1}\underline{p}$ $\quad \min_{\underline{w}} \frac{1}{L}\sum_{n=l_1}^{l_1+L-1} e^2[n] = \sigma_d^2 - \underline{p}^T\underline{R}^{-1}\underline{p}$

Connection to solving $\underline{d} \approx \underline{X}^T \underline{w}$

1) Multiply both sides by $\underline{X}$ (normal equations)

$$\underline{X}\,\underline{d} = \underline{X}\,\underline{X}^T \underline{w} \iff L\,\underline{p} = L\,\underline{R}\,\underline{w}$$

2) Invert $\underline{X}\,\underline{X}^T$:

$$\underline{w} = (\underline{X}\,\underline{X}^T)^{-1}\,\underline{X}\,\underline{d} \iff \underline{w} = \underline{R}^{-1}\,\underline{p}$$

Notes –

1) Existence of $\underline{R}^{-1}$ requires $L \geq N$. Noise usually ensures $\underline{R}^{-1}$ exists if $L \geq N$

2) If N is large or the data has wide dynamic range, it may be better to avoid computing $\underline{R}^{-1}$